

Technical Datasheet of WIN-IO-16UI8UOM-E Module (Modbus TCP/IP)

1. Overview

The Universal Input (UI) and Universal Output (UO) board support **communication exclusively over Modbus TCP/IP**, enabling seamless control and monitoring of inputs/outputs. The board provide **configurable input and output selection via TCP/IP web pages**, ensuring flexibility across various applications. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.

Digital IO module is sturdy, has low power usage and is easy to use.



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

Specifications

<u>Category</u>	<u>UIUO Module</u>
Input Type	Analog Input (10V) Analog Input (5V) Digital Input NTC Sensor Input. (Configurable via TCP/IP web page)
Output Types	Analog Output (AO), Digital Output (DO). (configurable via TCP/IP web pages)
Connectors	Single port handling multiple input/output types

General –

Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 9 – 24 V DC or 24 V AC/DC Typical – 24V DC @ 750ma
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



UI (Universal Inputs) –

Digital Input channels	16
Sense Voltage	3.3 – 12 VDC (Depends on the selected type of Input)
Digital Sensing Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec
Analog Input channels	16
Input Signal	0 – 5VDC / 0 – 10VDC / NTC (Web page selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits
Isolation	Optically Isolated
External Loop Voltage	+ 15 VDC min
AI input impedance	120Ω

UO (Universal Outputs) –

Digital Output channels	8
Output Type	P-Channel MOSFET
Channel Voltage	12 VDC
Channel Current	20A
Channel Power	240 W (max)
Operating Time	2.8ns.
Turn Off delay Time	33.5ns.
Operating Temperature	-55°C~+150°C

Analog Output channels	8
Analog Resolution	10 bits (0 to 10VDC)
Accuracy	± 2% Full scale
Isolation	Optically Isolated
Conversion Time	15 msec

Additional Features: -

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air

Configuration Of UIUO module

The UI/UO module supports flexible I/O configuration.

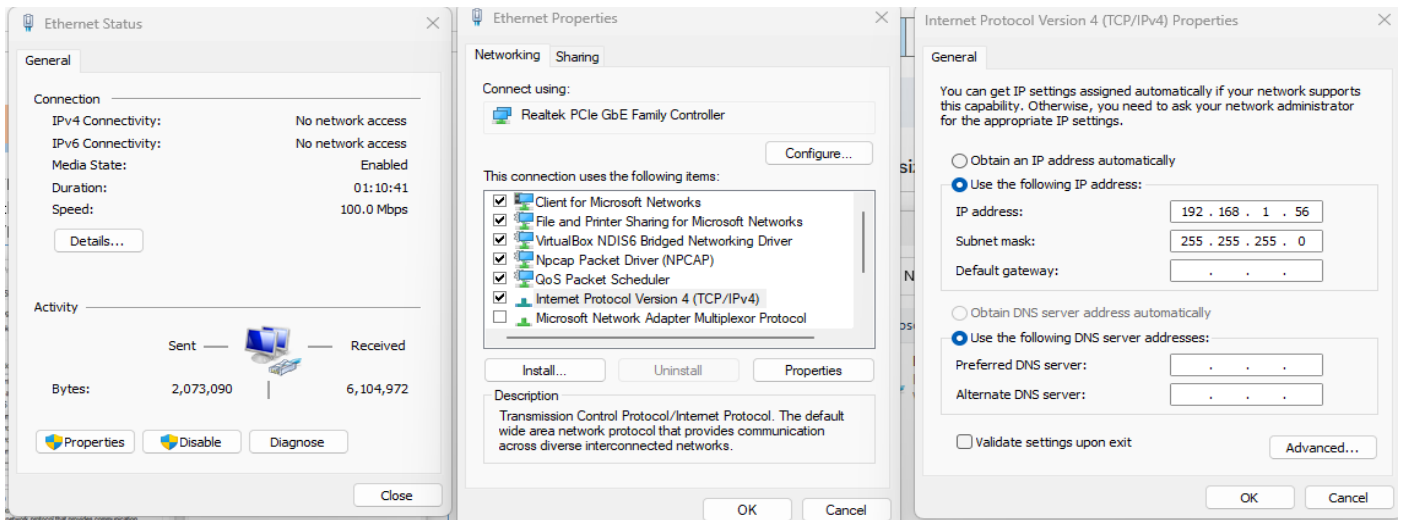
Each of the 16 UI (Universal Input) ports can be independently configured as Analog Input, NTC Sensor Input, Digital Input. Similarly, each of the 8 UO (Universal Output) ports can be independently configured as, Analog Output, Digital Output

All the configuration can be done through Web page hosted at the default IP address (i.e. 192.168.1.177) or at the changed IP address.

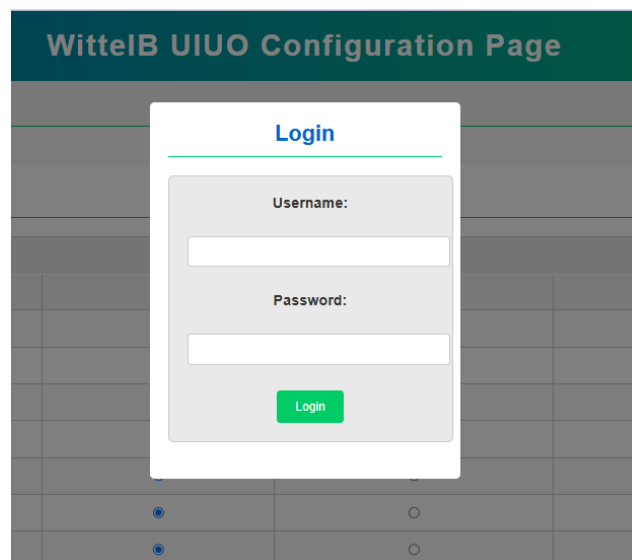
The following are step by step guide for configuration:

1. Connect the Module to +24VDC POWER SUPPLY.
2. Connect the Board to the PC via an Ethernet Cable.

3. Use the default setting for PC as TCP/IPv4 (i.e., Address -192.168.1.56).



4. Now, open web browser and type **192.168.1.177** (default IP address) and Press 'Enter'. A login page will appear as shown below.



5. Default Login Credentials are :

Username: admin

Password: admin

6. After successful login, The UIUO Configuration page will be visible. It has three tables. The first and second table named as 'Update UI Status' and 'Update UO Status' which have selectable radio button for doing the configuration of respective channels as shown below.

Update UI Status

CHANNEL	DI	10V	NTC	5V
1	☐	☒	☐	☐
2	☐	☐	☒	☐
3	☒	☐	☐	☐
4	☒	☐	☐	☐
5	☒	☐	☐	☐
6	☒	☐	☐	☐
7	☒	☐	☐	☐
8	☒	☐	☐	☐
9	☒	☐	☐	☐
10	☒	☐	☐	☐
11	☒	☐	☐	☐
12	☒	☐	☐	☐
13	☒	☐	☐	☐
14	☒	☐	☐	☐
15	☒	☐	☐	☐
16	☒	☐	☐	☐

Update UI Status

Update UO Status

CHANNEL	DO	AO
1	☐	☒
2	☐	☒
3	☐	☒
4	☐	☒
5	☐	☒
6	☐	☒
7	☐	☒
8	☐	☒

Update UO Status

7. For example, to set Channel 1 of UI as NTC Sensor input, the radio button in front of Channel 1 must be selected as NTC and then "Update UI Status" button need to be pressed. This will make the Channel 1 of UI port as NTC sensor input port.

Note: Always restart the device/refresh the page after doing configuration for UI/UO Status, then only changes in configuration will take place.

8. The third table “UI_DATA” is for viewing the UI port data locally on the web page.

Data Section

UI_DATA	
CHANNEL	UI_DATA
1	421
2	117
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0

9. User can select all the desired configuration at once and then press the Update button for UI or UO to update the Port status in the module.
10. User can change the IP and Mac Address for the TCP from the second tab “MODBUS TCP CONFIGURATION” in the web page and press the “Save Configuration” button and restart the device to set the new IP, as shown below.

UIUO PAGE
MODBUS TCP CONFIGURATION
UPDATE LOGIN

Modbus TCP Configuration

MAC ID:

F8:B3:B7:A5:2F:2B

IP Address:

192.168.1.177

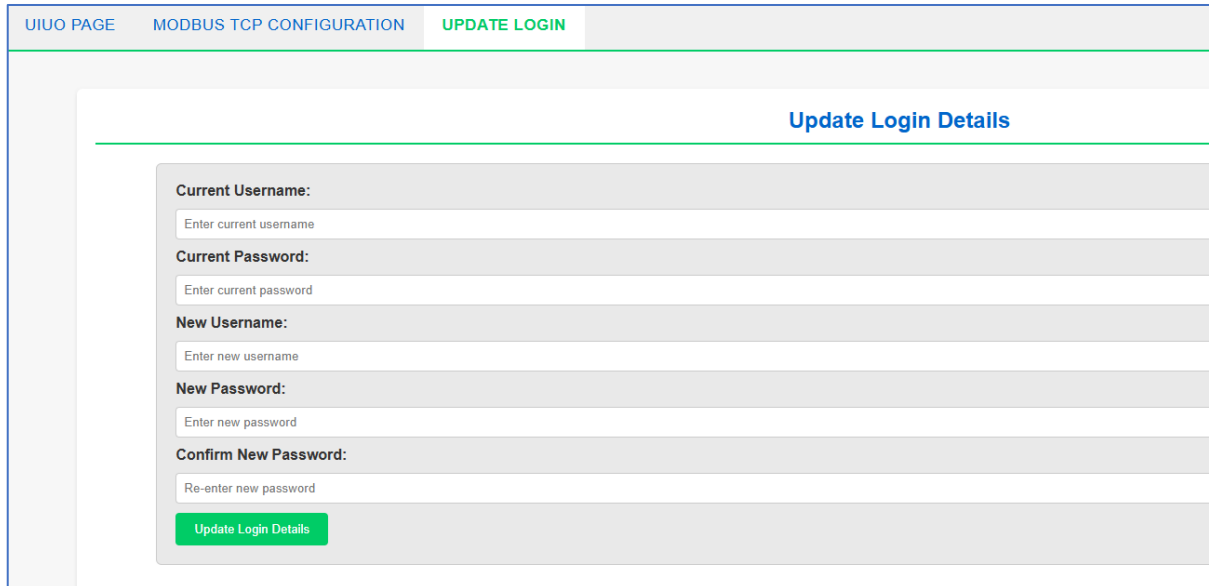
Port:

502

Save Configuration

Note: To reset the IP address to default IP, the user need to press and hold the reset button located near the Ethernet port for 5 seconds.

11. On the third tab “UPDATE LOGIN” page, the user can change the Username and Password, to enhance the security of UIUO module, as show below.



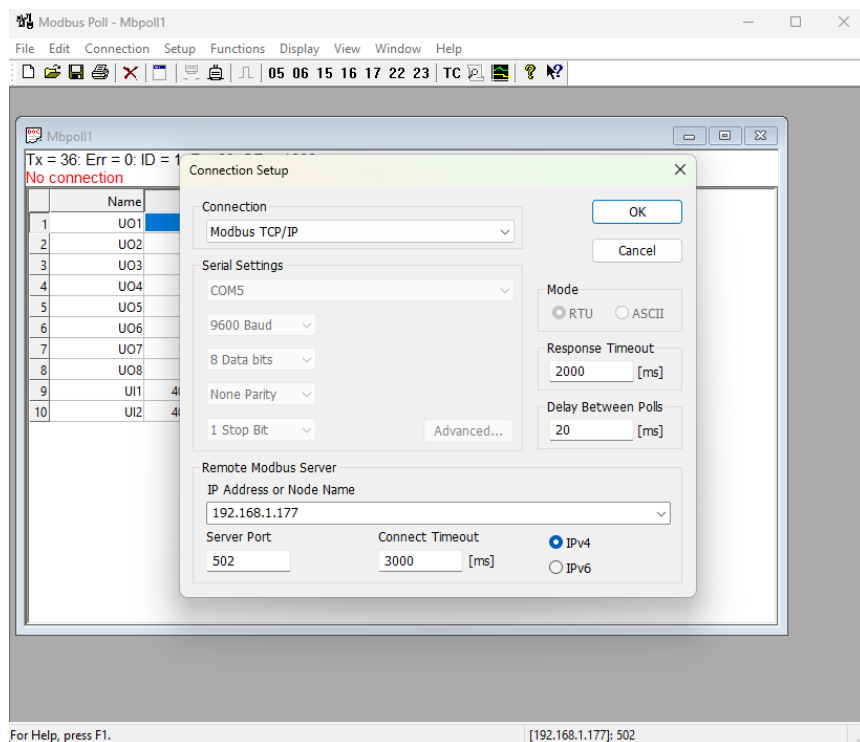
The screenshot shows the 'UPDATE LOGIN' tab selected in the top navigation bar. The main content area is titled 'Update Login Details'. It contains the following fields:

- Current Username:** A text input field with the placeholder 'Enter current username'.
- Current Password:** A text input field with the placeholder 'Enter current password'.
- New Username:** A text input field with the placeholder 'Enter new username'.
- New Password:** A text input field with the placeholder 'Enter new password'.
- Confirm New Password:** A text input field with the placeholder 'Re-enter new password'.

At the bottom of the form is a green button labeled 'Update Login Details'.

12. After all the configuration, User can now connect the UIUO module to Modbus TCP master at the set IP address or default IP address (192.168.1.177).

13. Here we have shown a connecting with Modbus Poll software for demonstration in the below images.



Mbpoll1

Tx = 40: Err = 0: ID = 1: F = 03: SR = 1000ms

	Name	4x0000	Name	4x0010	Name	4x0020
1	UO1	40001 = 0	UI3	40011 = 0	UI13	40021 = 0
2	UO2	40002 = 0	UI4	40012 = 0	UI14	40022 = 0
3	UO3	40003 = 0	UI5	40013 = 0	UI15	40023 = 0
4	UO4	40004 = 0	UI6	40014 = 0	UI16	40024 = 0
5	UO5	40005 = 0	UI7	40015 = 0		
6	UO6	40006 = 0	UI8	40016 = 0		
7	UO7	40007 = 0	UI9	40017 = 0		
8	UO8	40008 = 0	UI10	40018 = 0		
9	UI1	40009 = 418	UI11	40019 = 0		
10	UI2	40010 = 117	UI12	40020 = 0		

For Help, press F1. [192.168.1.177]: 502

14. The data of UI port can be fetched from the mentioned Modbus register in the table below, irrespective of DI, AI, NTC, selection. The Desired Inputs are as follows:

INPUT TYPE	Input Value
DI	0 V(0) , 12VDC (1)
AI(5V)	0 to 5VDC (0-1023)
AI(10V)	0 to 10VDC (0-1023)
NTC	NTC sensor(Temperature in °C)

15. Similarly, the data of the UO port can be set in the same register as mentioned in the table below. For DO, It should be sent 0 or 1 (12VDC). And for AO a 10-bit resolution value must be sent (i.e. from 0 to 1023) to set output of 0 to 10VDC.

OUTPUT Type	Output Value
DO	0 (0V) , 1 (12VDC)
AO	0 to 10VDC (0-1023)

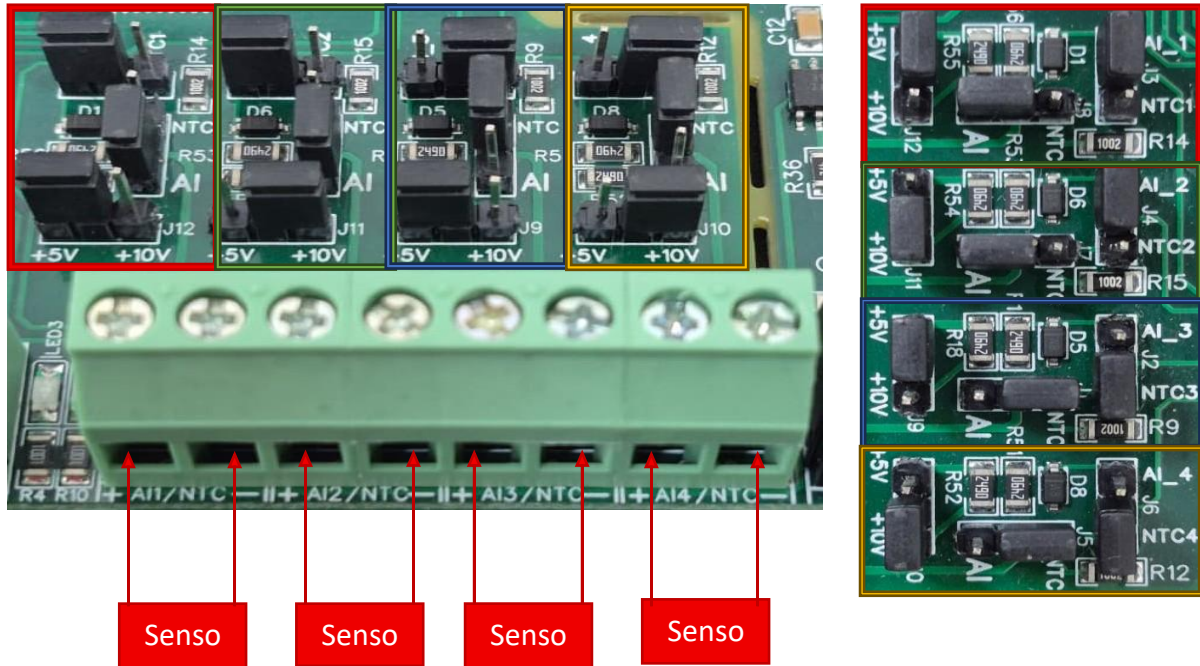
MODBUS COILS TABLE FOR UIUO PORTS:-

ID	Function Description	Register Description	Modbus Function Code	Data Type
1	UO 1 (DO)	00001	0x05,0x15,0x01	Boolean
2	UO 2 (DO)	00002	0x05,0x15,0x01	Boolean
3	UO 3 (DO)	00003	0x05,0x15,0x01	Boolean
4	UO 4 (DO)	00004	0x05,0x15,0x01	Boolean
5	UO 5 (DO)	00005	0x05,0x15,0x01	Boolean
6	UO 6 (DO)	00006	0x05,0x15,0x01	Boolean
7	UO 7 (DO)	00007	0x05,0x15,0x01	Boolean
8	UO 8 (DO)	00008	0x05,0x15,0x01	Boolean
9	UI 1 (DI)	00009	0x01	Boolean
10	UI 2 (DI)	00010	0x01	Boolean
11	UI 3 (DI)	00011	0x01	Boolean
12	UI 4 (DI)	00012	0x01	Boolean
13	UI 5 (DI)	00013	0x01	Boolean
14	UI 6 (DI)	00014	0x01	Boolean
15	UI 7 (DI)	00015	0x01	Boolean
16	UI 8 (DI)	00016	0x01	Boolean
17	UI 9 (DI)	00017	0x01	Boolean
18	UI 10 (DI)	00018	0x01	Boolean
19	UI 11 (DI)	00019	0x01	Boolean
20	UI 12 (DI)	00020	0x01	Boolean
21	UI 13 (DI)	00021	0x01	Boolean
22	UI 14 (DI)	00022	0x01	Boolean
23	UI 15 (DI)	00023	0x01	Boolean
24	UI 16 (DI)	00024	0x01	Boolean

MODBUS REGISTERS TABLE FOR UIUO PORTS:-

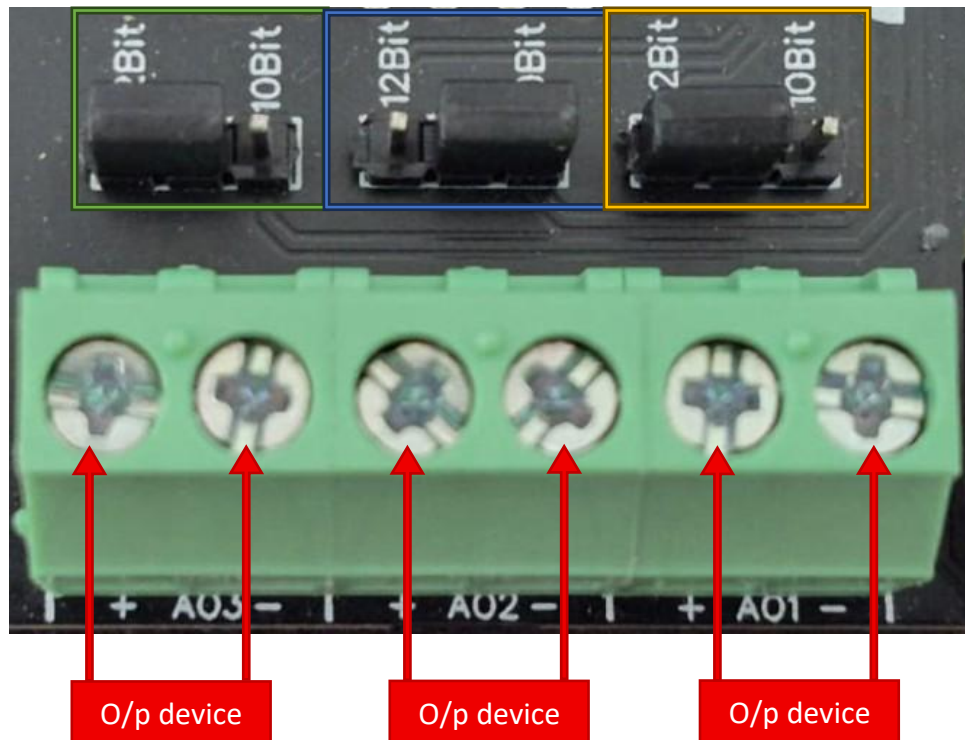
1	UO 1 (AO)	40001	0X03	16 Bit Unsigned int
2	UO 2 (AO)	40002	0X03	16 Bit Unsigned int
3	UO 3 (AO)	40003	0X03	16 Bit Unsigned int
4	UO 4 (AO)	40004	0X03	16 Bit Unsigned int
5	UO 5 (AO)	40005	0X03	16 Bit Unsigned int
6	UO 6 (AO)	40006	0X03	16 Bit Unsigned int
7	UO 7 (AO)	40007	0X03	16 Bit Unsigned int
9	UI 1 (AI,NTC)	40009	0x03	16 Bit Unsigned int
10	UI 2 (AI,NTC)	40010	0x03	16 Bit Unsigned int
11	UI 3 (AI,NTC)	40011	0x03	16 Bit Unsigned int
12	UI 4 (AI,NTC)	40012	0x03	16 Bit Unsigned int
13	UI 5 (AI,NTC)	40013	0X03	16 Bit Unsigned int
14	UI 6 (AI,NTC)	40014	0X03	16 Bit Unsigned int
15	UI 7 (AI,NTC)	40015	0X03	16 Bit Unsigned int
16	UI 8 (AI,NTC)	40016	0X03	16 Bit Unsigned int

AI-NTC



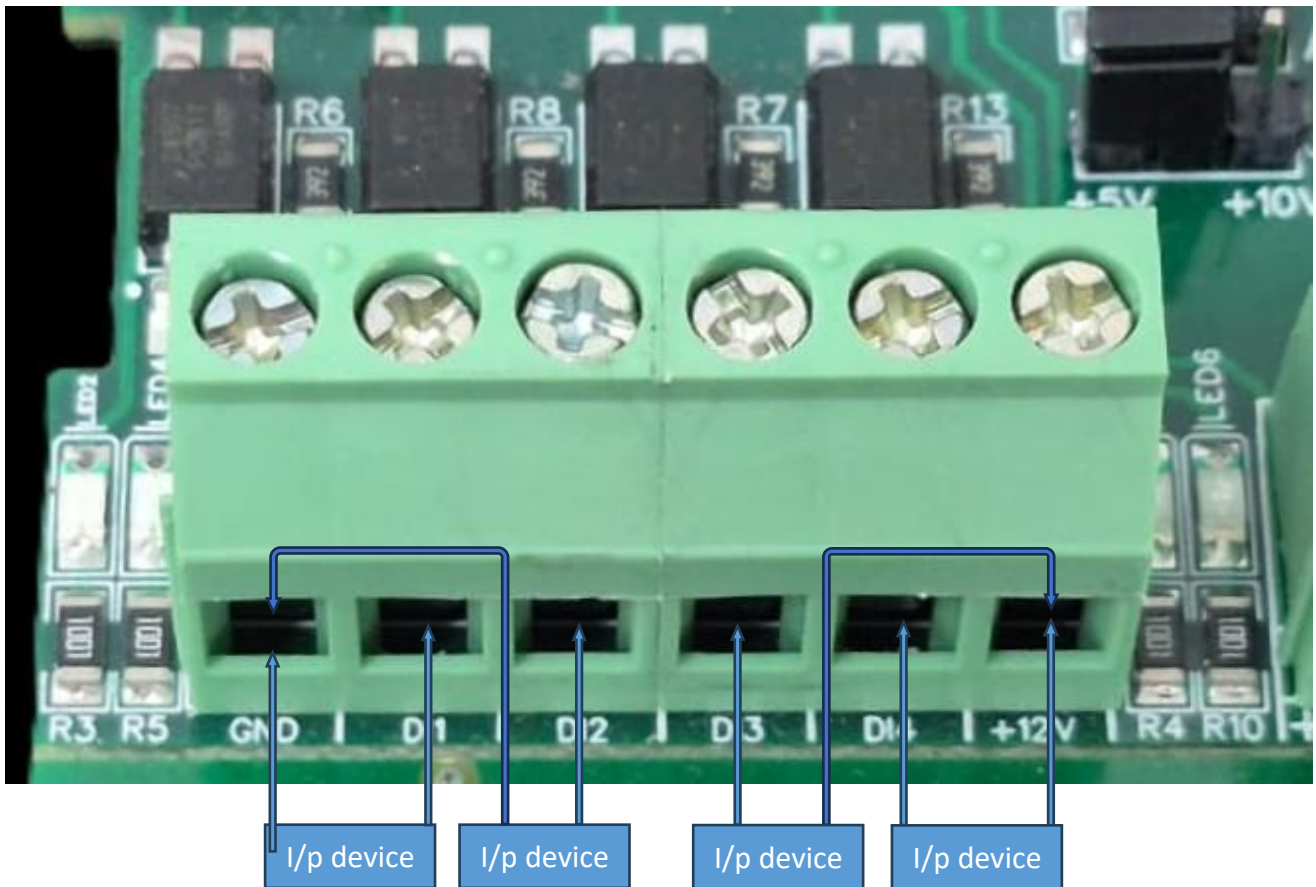
- **Analog Input & NTC Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to interface with either standard external analog sensors or NTC thermistors.
- **Channel Configuration (Visual Guide):** Each of the four channels features two dedicated jumpers to independently select the operating mode (AI or NTC) and the reference voltage range (5V or 10V). Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (AI Mode, 5V Range):** Set the mode jumper to **AI** and the voltage jumper to **+5V**.
 - **Green Box (AI Mode, 10V Range):** Set the mode jumper to **AI** and the voltage jumper to **+10V**.
 - **Blue Box (NTC Mode, 5V Range):** Set the mode jumper to **NTC** and the voltage jumper to **+5V**.
 - **Yellow Box (NTC Mode, 10V Range):** Set the mode jumper to **NTC** and the voltage jumper to **+10V**.
- **Sensor Wiring Connections:**
 - The sensor or thermistor's positive signal wire should be connected to the + terminal of the desired channel (labelled AI1/NTC, AI2/NTC, etc.).
 - The sensor's common or ground wire should be connected to the corresponding - terminal.
- **Crucial Setup Recommendation:** Always ensure that both the mode jumper (NTC vs. AI) and the voltage jumper strictly match the hardware you are connecting. Incorrectly setting an NTC to an AI mode, or mismatching the voltage limits, will result in inaccurate temperature or signal readings.

AO



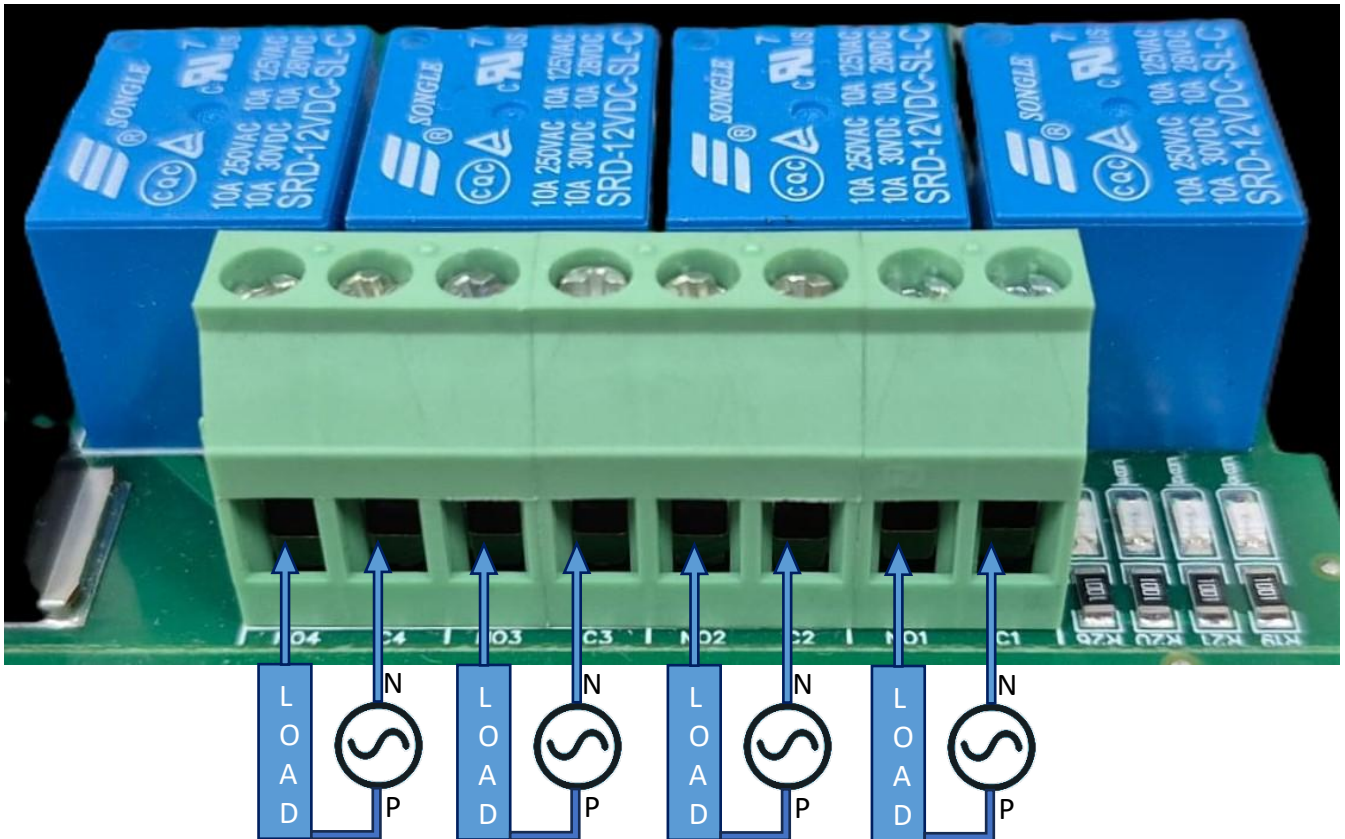
- **Analog Output (AO) Operation:** Each channel provides two terminals: + (**Positive**) and - (**Negative/Common**). These are used to send variable analog signals to **external output (O/p) devices**, such as variable frequency drives or proportional valves.
- **Channel Configuration (Visual Guide):** Each of the three AO channels features a **dedicated jumper** to set the **output bit resolution**. Refer to the color-coded boxes in the board diagram to configure your desired resolution:
 - **Green & Yellow Box (12-bit):** Set the resolution jumper to **12Bit**.
 - **Blue Box (10-bit):** Set the resolution jumper to **10Bit**.
- **Output Device Wiring Connections:**
 - The external device's positive signal wire should be connected to the + terminal of the desired channel (AO1, AO2, or AO3).
 - The device's common or ground wire should be connected to the corresponding - terminal..
- **Crucial Setup Recommendation:** Always ensure that the hardware jumper setting (10-bit or 12-bit) matches the output scaling configured in your main control software. A mismatch between software resolution and hardware configuration will result in incorrect output signal levels.

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

DO



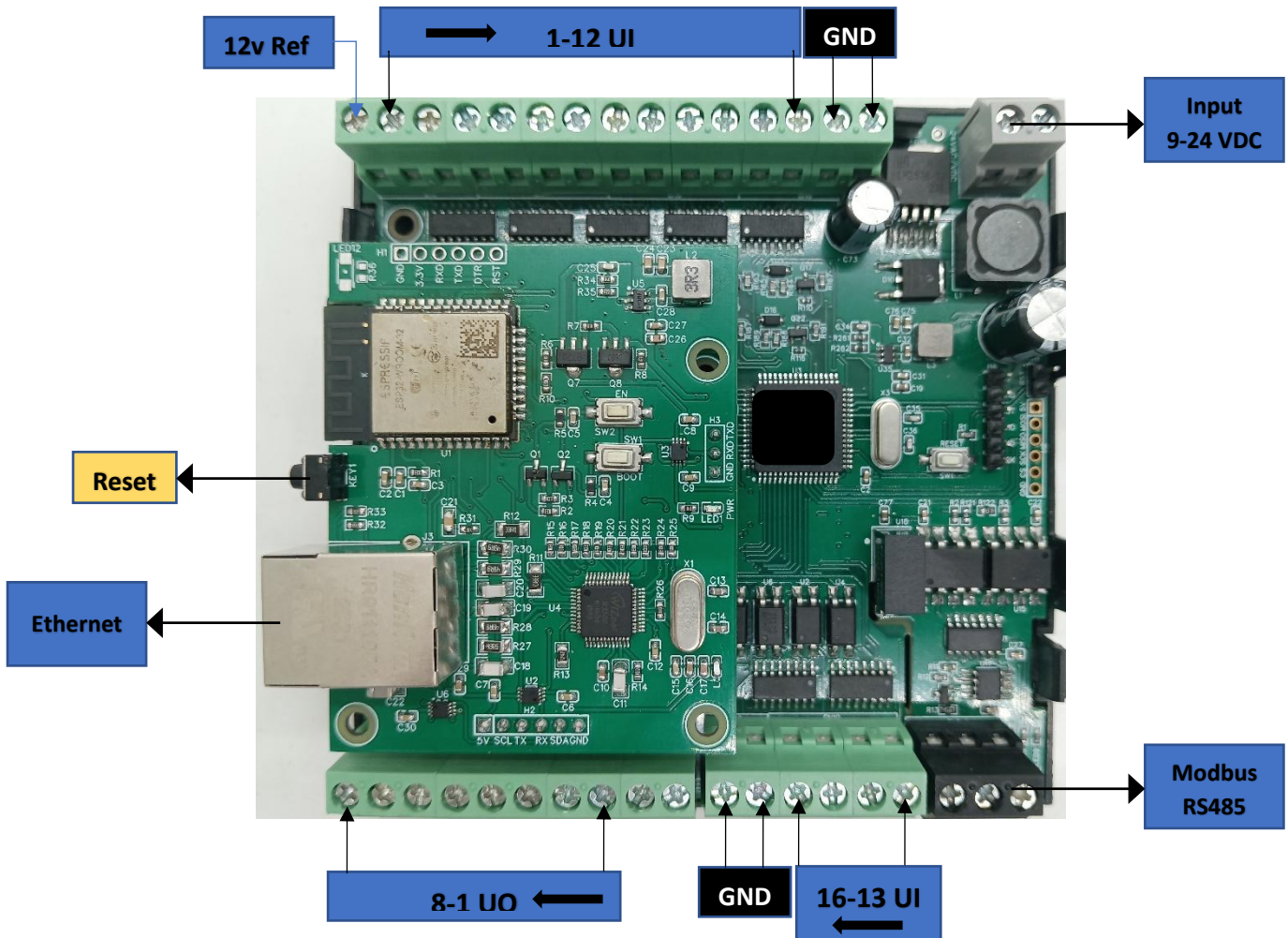
Digital Output (DO) operation: Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C** and **NO** **remains open**, and the load stays OFF.

- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

Wiring Diagram:-

The Port number and Sequence of UI and UO are as mentioned in the wiring diagrams shown below. Also the position of default reset button is as shown in the diagram.



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